

Capstone Project

Deep Learning

Final Submission

MIT Applied AI and Data Science Program

Malaria Detection

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Executive Summary

Malaria is a serious global public health problem and diagnosis traditionally depends on microscopic review of blood smear images by trained professionals. While effective, this process can be time-consuming, labor-intensive, and subject to variability across reviewers.

This project developed a deep learning solution to classify red blood cell images as **parasitized** or **uninfected** to support automated malaria screening. The dataset was well balanced across both classes, and visual exploration showed that infection-related features were often subtle, localized, and related to staining intensity, color variation, texture, and small intracellular structures. These findings supported the use of convolutional neural networks for image-based classification.

Several models were evaluated, including a baseline CNN, modified CNN architectures, a CNN with data augmentation, a VGG16 transfer-learning model, and an HSV-based CNN. The HSV CNN was selected as the final model because it provided the best balance between overall accuracy and detection of parasitized cells. It achieved approximately **98.35% test accuracy**, correctly identified **1,284 out of 1,300 parasitized images**, and produced only **16 false negatives**, corresponding to a parasitized recall of approximately **98.77%**.

The final proposed solution is an HSV-based convolutional neural network that uses color and spatial information to detect subtle staining and structural patterns associated with parasitized cells. The architecture was designed to progressively extract increasingly complex visual features while controlling overfitting.

The proposed model should be implemented as a decision-support tool for malaria image screening. It can help prioritize potentially infected cells for expert review, improve consistency, and support faster screening workflows. Before deployment, the model should be validated on external datasets collected under different staining, imaging, and laboratory conditions.

Final Proposed Model Specifications

Component	Final Specification
Final model	HSV-based Convolutional Neural Network
Task	Binary image classification
Input data	Red blood cell images
Input size	$64 \times 64 \times 3$
Preprocessing	Normalization to 0–1 scale and RGB-to-HSV conversion
Output classes	Uninfected and Parasitized
Architecture	Three convolutional blocks, max pooling, dropout, dense layer, softmax output
Output layer	Dense layer with 2 units and softmax activation
Key metric	Recall for the parasitized class
Test accuracy	~98.35%
Parasitized recall	~98.77%
False negatives	16
Recommended use	Decision-support tool for malaria screening

Problem Summary

Malaria remains a major global public health challenge caused by Plasmodium parasites that infect and damage red blood cells. The disease can lead to severe complications and death, particularly among vulnerable populations such as children under five years of age. Because malaria infection may initially be asymptomatic, delayed diagnosis can increase the risk of disease progression and continued transmission.

Manual malaria screening from blood smear images requires trained personnel and careful microscopic review. This can create operational challenges when image volume is high, expert availability is limited, or consistency across reviewers is difficult to maintain. The key business problem is therefore not only classification accuracy, but also the need for a scalable and reliable screening-support process that can reduce missed infected cells.

Solution summary

The proposed solution is an HSV-based convolutional neural network that classifies red blood cell images as **parasitized** or **uninfected**. This solution was selected after a careful comparison of several deep learning approaches, including a baseline CNN, modified CNN architectures, a CNN with image augmentation, a VGG16 transfer-learning model, and an HSV-based CNN. Among the evaluated models, the HSV CNN achieved the best overall performance for the malaria detection objective.

Before classification, images were resized to **64 × 64 × 3**, normalized to a **0–1 pixel range**, and converted from RGB to **HSV color space**. This transformation separates hue, saturation, and brightness information and may help distinguish subtle staining and color patterns associated with malaria infection. The CNN then progressively extracts increasingly complex visual features through three convolutional blocks with **32, 64, and 128 filters**, followed by pooling and dropout regularization to reduce overfitting.

The model was trained using a fixed **train-validation-test strategy**. Twenty percent of the training data was reserved for validation using a stratified split to preserve class balance, while the separate test set was kept untouched for final evaluation. Image augmentation was applied only to the training data so that validation and test performance reflected the model's ability to generalize to unseen, unmodified images.

The training and validation curves show that model performance improved during training, with accuracy increasing and loss decreasing over successive epochs. The relatively close behavior of the training and validation curves suggests that the model generalized well without evidence of severe overfitting (Fig. 1). Early stopping and dropout regularization were used to reduce overfitting and retain the best-performing model.

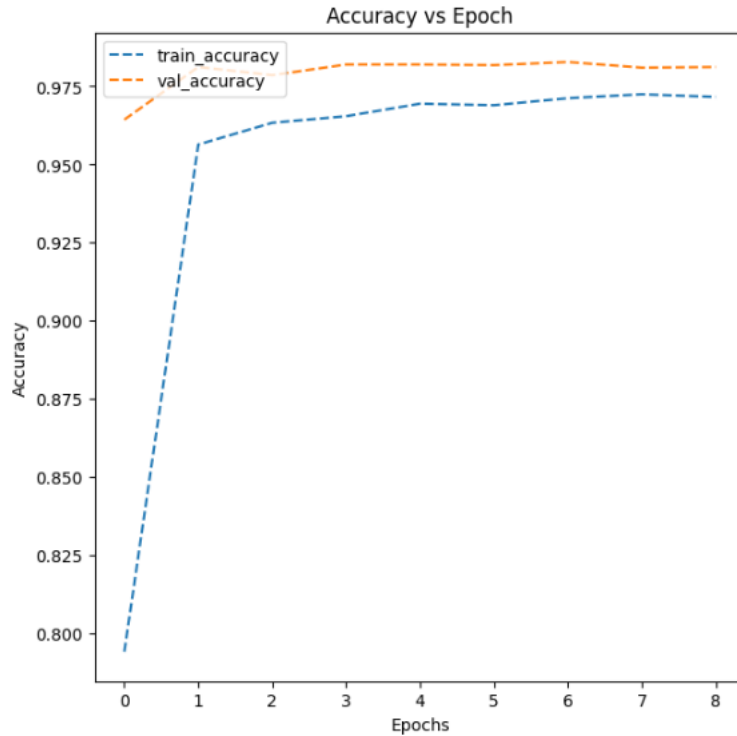


Figure 2. Training and validation accuracy curves for the HSV CNN Model.

On the test set, the HSV CNN achieved approximately **98.35% accuracy** and correctly identified **1,284 of 1,300 parasitized images**, corresponding to a parasitized recall of approximately **98.77%**. The model produced only **16 false negatives**, the lowest false-negative count among the evaluated approaches (Fig. 2). This result is particularly important for the screening objective because false negatives represent infected cells that the model fails to identify.

The confusion matrix also shows **27 false positives**, representing uninfected images incorrectly flagged as parasitized. These cases would create additional expert-review workload, but they are operationally less concerning than false negatives because they do not represent missed infected cells. In a decision-support workflow, this trade-off may be acceptable when the priority is to maximize sensitivity and reduce the risk of overlooking potentially infected cells.

Overall, the results suggest that the model could serve as an effective **first-pass screening and triage tool**. Rather than requiring experts to review every image with equal priority, the model could help identify images more likely to contain parasitized cells and direct expert attention accordingly. This has the potential to reduce manual screening burden, improve consistency, and support faster review of large image volumes.

However, the reported results represent **image-level classification performance on the available test dataset**, not patient-level diagnostic accuracy. Therefore, the model should not be interpreted as a standalone diagnostic system. **The appropriate next step is external validation**

on more diverse datasets, followed by controlled pilot testing to determine whether the observed performance translates into measurable workflow improvements under real-world laboratory conditions.

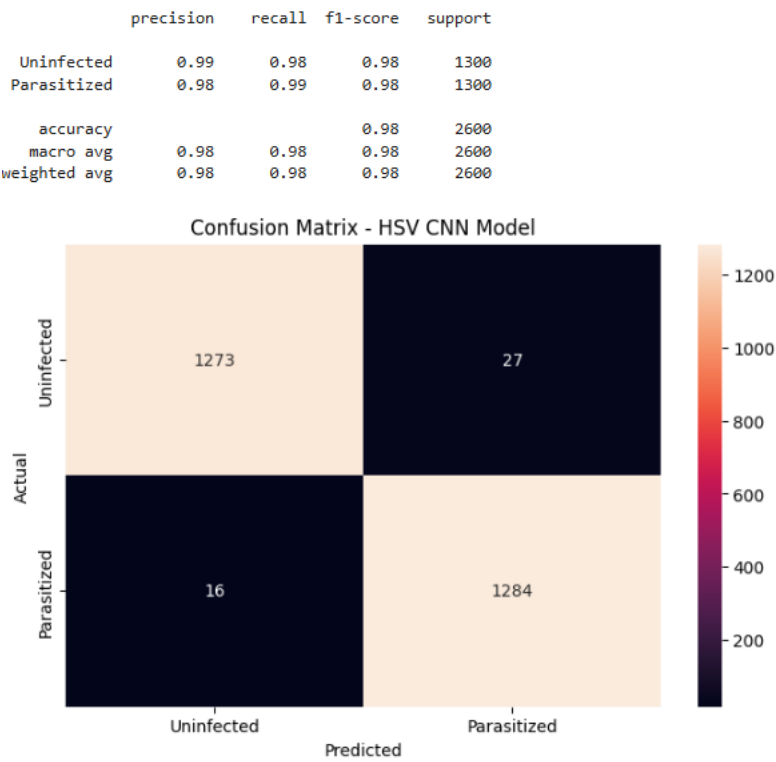


Figure 2. HSV CNN model classification report and confusion matrix.

Recommendations for Implementation

●Use the Model as a Decision support-tool

The HSV CNN model should be implemented as a **decision-support tool** for malaria image screening rather than as a replacement for expert review. The model should classify red blood cell images as **parasitized** or **uninfected**, with parasitized predictions flagged for review by trained laboratory personnel.

●Validate the Model on External Data

Before implementation, the model should be validated on external datasets collected under different staining protocols, microscopes, image resolutions, laboratory conditions, and patient populations. External validation is essential to determine whether the model maintains its strong performance beyond the development dataset.

●Optimize the Operating Threshold

The classification threshold should be optimized around the main screening objective: **maximizing parasitized recall and minimizing false negatives**. A more sensitive threshold may identify more infected cells but also increase false positives and expert-review workload. The final threshold should therefore balance missed-infection risk with operational capacity.

●Implement in Phases and Monitor Performance

Deployment should begin with a controlled pilot before broader implementation. The pilot should measure screening time, false-negative rate, reviewer workload, turnaround time, and user acceptance. If the model demonstrates consistent performance and measurable workflow benefits, implementation can then be expanded gradually.

Once deployed, performance should be monitored using **accuracy, precision, recall, F1-score, confusion matrices, and especially false-negative counts**. Meaningful declines in recall, or changes in image quality, staining, microscope settings, or patient population, should trigger review, recalibration, or retraining. This monitoring process is essential to maintain reliability, safety, and operational value over time.

Key Actions for Stakeholders

The **data science team** should standardize the preprocessing pipeline: image resizing to **64 × 64 × 3**, pixel normalization, RGB-to-HSV conversion, and classification using the trained CNN model. The team should also monitor model performance over time using accuracy, precision, recall, F1-score, confusion matrices, and false-negative counts. Special attention should be given to **parasitized recall** and **false negatives**, because false negatives represent infected cells that the model fails to identify. Monitoring should also include periodic review of incorrectly classified images to determine whether errors are related to staining quality, image artifacts, microscope differences, or limitations of the model itself.

Laboratory and clinical stakeholders should define how model outputs will be used in practice. For example, images classified as parasitized could be prioritized for expert review, while images classified as uninfected could remain subject to quality-control checks. This workflow should improve efficiency while maintaining expert accountability.

The **operational leadership team** should evaluate the infrastructure needed to deploy and maintain the model. This includes image storage, secure data transfer, model hosting, integration with microscopy or digital pathology workflows, reporting interfaces, user access controls, staff training, and ongoing technical support. Leaders should also define ownership of the model after deployment, including who is responsible for performance monitoring, retraining, validation, documentation, and approval of future updates.

Operational leaders should also plan for a pilot implementation before broader deployment. During the pilot, the organization should measure workflow impact, including screening time, number of images flagged for expert review, reviewer workload, false-negative rate, turnaround time, and user acceptance. These results would help determine whether the model provides measurable operational value and whether adjustments are needed before scaling.

Expected Business and Operational Impact

The proposed HSV CNN solution could support a more scalable, consistent, and efficient malaria screening workflow. By automatically classifying red blood cell images as **parasitized** or **uninfected**, the model can help prioritize images that are more likely to contain infected cells for expert review. This can reduce the time required to screen large image volumes and allow trained personnel to focus their attention on cases with higher diagnostic relevance.

The model's strong test performance, particularly its **98.77% recall for the parasitized class**, suggests that it could provide meaningful operational value while keeping missed infected cells relatively low.

From a business and operational perspective, the model **could improve workflow efficiency by reducing manual workload, supporting faster image triage, and improving consistency across reviewers**. It could **also strengthen quality-control processes** by providing a standardized first-pass screening method that applies the same decision rules across all images. This may be particularly valuable in high-volume laboratory settings or in environments where expert microscopy resources are limited.

The solution may also help **improve resource allocation**. Instead of requiring experts to review every image with equal priority, the model can help direct attention toward images most likely to require confirmation. This can support better use of staff time, faster turnaround, and more consistent screening performance. However, the model should be implemented as a **decision-support system**, not as a standalone diagnostic tool. Expert review and external validation remain necessary before real-world deployment.

Key Risks and Challenges

The main risk is **limited generalization to new data**. Images collected from different laboratories, microscopes, staining procedures, or patient populations may differ from the training data. If these differences are substantial, model performance may decline. This risk should be addressed through external validation and ongoing performance monitoring.

Another risk is **overreliance on AI-model predictions**. The model should not be used as a standalone diagnostic system without expert confirmation. False negatives remain clinically important, and false positives may increase review workload. Therefore, model outputs should be integrated into a controlled decision-support workflow with human oversight.

Additional challenges include maintaining image quality, ensuring consistent preprocessing, monitoring model drift, and defining responsibility for reviewing flagged cases. The organization would also need **clear governance for data privacy, model ownership, version control, performance monitoring, approval of model updates, and clinical validation** before real-world deployment.

Further Analysis Needed

Benchmark Against Alternative Architectures

Further analysis should compare the HSV CNN with recent CNN and transfer-learning approaches, including architectures such as **ResNet, EfficientNet, MobileNet, DenseNet, Inception**, and other microscopy-specific models. The comparison should consider not only overall accuracy, but also **parasitized recall, false-negative rate, model complexity, inference speed, interpretability, and ease of deployment**.

This analysis would help determine whether the HSV CNN remains the strongest candidate or whether alternative architectures could provide better detection performance or operational advantages.

Explore Alternative Preprocessing and Model Configurations

Additional model development should evaluate whether performance can be improved through alternative preprocessing and training strategies. Potential areas include **different image sizes, color spaces, contrast-enhancement methods, stain normalization techniques, and augmentation strategies**.

Because malaria-related visual features can be subtle, these experiments should determine whether alternative preprocessing methods improve detection of parasitized cells while maintaining a low false-negative rate.

Improve Model Robustness and Stability

Future work could use **K-fold cross-validation** or repeated validation splits, if computational resources permit, to determine whether performance remains stable across different subsets of the data. This would provide greater confidence that the reported results are not overly dependent on a single train-validation split.

Optimize Model Hyperparameters

Systematic hyperparameter tuning should evaluate settings such as **learning rate, dropout rate, batch size, number of convolutional filters, dense-layer size, augmentation parameters, and training duration**. These experiments should assess whether alternative configurations improve parasitized recall, reduce false negatives, strengthen model stability, or lower computational requirements.

A structured optimization strategy, such as random search or Bayesian optimization, could also be considered to identify promising configurations more efficiently than manual trial-and-error testing.

Conduct Detailed Error Analysis

Misclassified images, particularly **false negatives**, should be reviewed systematically to identify recurring failure patterns. Errors may be associated with poor image quality, staining artifacts, low parasite visibility, overlapping cells, unusual morphology, or limitations in the model's learned representations.

Understanding these failure modes could help determine whether future improvements should focus on preprocessing, data quality, architecture design, or additional training examples.

Evaluate Model Interpretability and Confidence

Further analysis should investigate how confident the model is in its predictions and which image regions contribute most strongly to classification decisions. Interpretability techniques could help determine whether the CNN is focusing on biologically meaningful parasite-related features rather than irrelevant background patterns or staining artifacts.

Confidence calibration could also help identify uncertain cases that may require additional expert review.

Overall Analytical Priority

The main objective of further analysis should be to determine whether the HSV CNN's strong performance is **reproducible, technically justified, robust, and explainable**. The highest priorities are therefore benchmarking alternative architectures, testing additional preprocessing strategies, optimizing hyperparameters, assessing stability across data splits, analyzing classification errors, and improving interpretability.